

Revealing the Environmental Advantages of Industrial Symbiosis in Wood-Based Bioeconomy Networks

An Assessment From a Life Cycle Perspective

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Summary

The German government has recently initiated funding schemes that incentivize strategies for wood-based bioeconomy regions. Regional wood and chemical industries have been encouraged to act symbiotically, that is, share pilot plant facilities, couple processes where feasible, and cascade woody feedstock throughout their process networks. However, during the planning stages of these bioeconomy regions, options need to be assessed for sustainably integrating processes and energy integration between the various industries that produce bio-based polymers and engineered wood products. The aim of this paper is to identify the environmental sustainability of industrial symbiosis for producing high-value-added, bio-based products in the wood-based bioeconomy region of Central Germany. An analysis was conducted of three possible future scenarios with varying degrees of symbiosis in the bioeconomy network. A life cycle assessment (LCA) approach was used to compare these three scenarios to a traditional fossil-based production system. Eleven environmental impact categories were considered. The results show that, in most cases, the bioeconomy network outperformed the fossil-based production system, mitigating environmental impacts by 25% to 130%.

Introduction

In recent years, the German government has promoted cluster initiatives for the development of sustainable wood-based bioeconomy regions (BMELV-Referat 531 2013). In these bio-based regions, shared access to pilot plants has also been promoted as a useful strategy for strengthening ties among industrial cooperation partners. This has opened up opportunities for testing process scalability (i.e., upscaling) and for evaluating the sustainability of biorefinery facilities

for intermediate production (SmartPilots Interreg Europe 2017; Winther and Klarlund 2014; Clusterplattform Germany Federal Ministry for Economic Affairs and Energy 2017). Another successful strategy has been the promotion of industrial symbiosis (IS), that is, industries cooperating in developing novel processing routes for by-products (Realf and Abbas 2003; Biobased Industries Consortium 2016). In combination, these strategies are regarded as the cornerstones of successful implementation of the bioeconomy and circular economy

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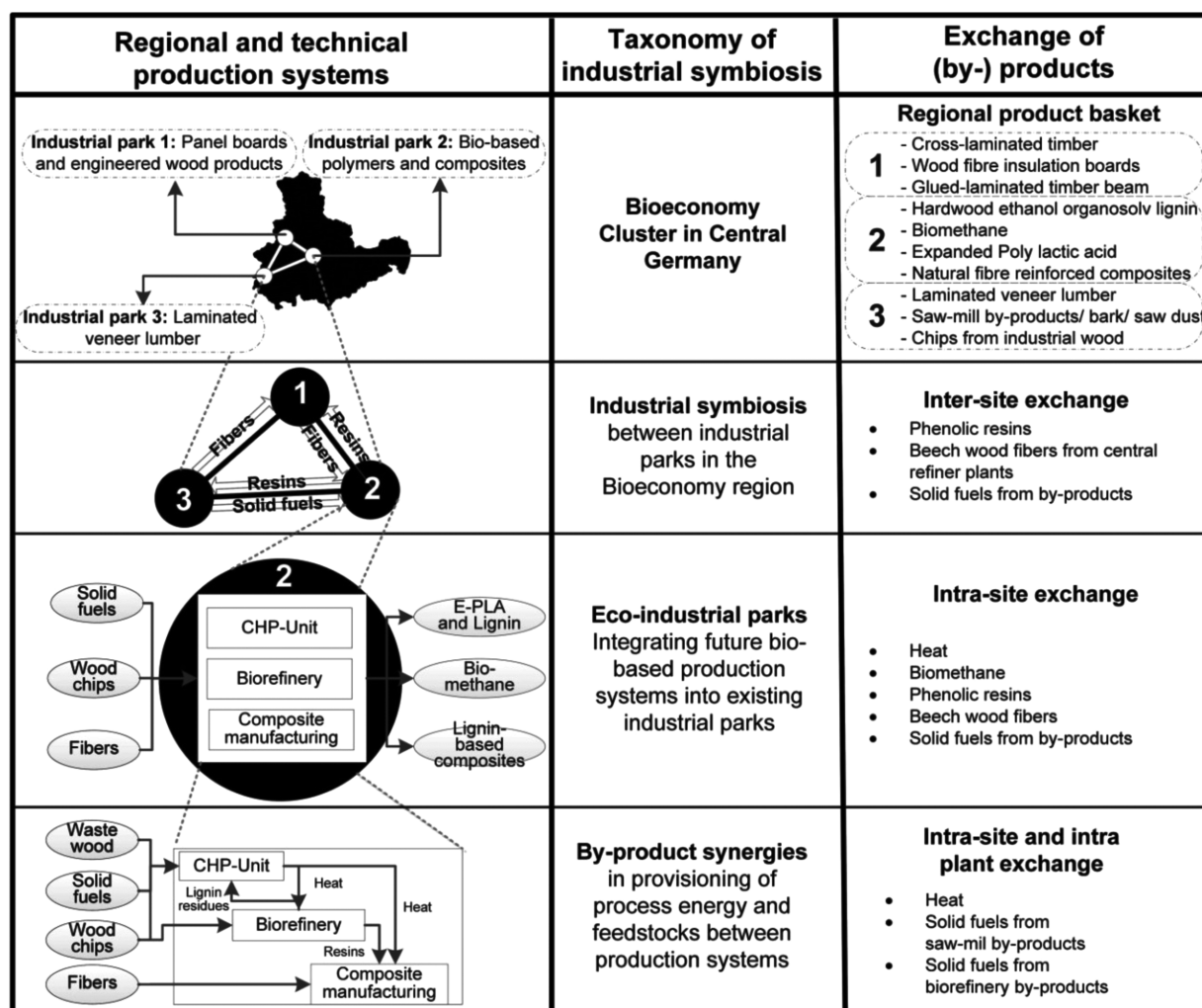


Figure 1 Taxonomic levels of industrial symbiosis in the regional bio-based production systems.

(Berlina et al. 2016; Biobased Industries Consortium 2016; BMELV-Referat 531 2013; Realff and Abbas 2003). There are two major ways in which these strategies can be practically implemented. The first involves options for adapting or retrofitting existing production infrastructures and networks, such as the ecological modernization of conventional refinery sites into “eco-industrial parks”(VDI 2014). In such parks, synergies can be identified among biomass suppliers, biotechnology companies, chemical companies, and waste facility operators, which facilitate the implementation of IS (Røyne et al. 2017; Sokka et al. 2011). The second relates to the development of new pilot and demonstration plant facilities that are supported by government funding initiatives and are run synergistically by different industries (e.g., pilot plants at EW Biotech and Fraunhofer CBP Leuna) (SmartPilots Interreg Europe 2017; BMELV-Referat 531 2013; VDI 2014). These are then used to showcase the advantages of IS and to encourage industrial cooperation among different regional industries (Berlina et al. 2016). This catalyzes traditional linear production systems toward more integrated regional bioeconomy networks (SmartPilots Interreg Europe 2017; Winther and

Klarlund 2014) and enables a more efficient utilization of wood resources, including the cascade use of waste wood and bio-based polymers (Hildebrandt et al. 2017a; Bezama 2016). It also promotes the more intensified use of wood resources for high added-value applications, in particular building materials (Hildebrandt et al. 2017c). In essence, these strategies have to be successfully combined in order to ensure the robust and sustainable development of bioeconomy regions (Thrän and Bezama 2017). The BioEconomy Cluster in Central Germany (SCBE) is one example of the development and testing of shared pilot facilities and innovative valorization options for lignocellulosic by-products. Established in 2012, it makes use of the associated piloting infrastructure of Fraunhofer CBP at the Leuna refinery site (SmartPilots Interreg Europe 2017; BMELV-Referat 531 2013; Clusterplattform Germany Federal Ministry for Economic Affairs and Energy 2017). The SCBE was launched with government support to facilitate cross-sectoral cooperation, research and development (R&D) pipeline management, and corporate development for wood-based added-value chains. The shared operation of pilot plants was a key element within the cluster for advancing bio-based

processing routes and IS among the various companies involved. Increasing the options for IS from a systems perspective within the SCBE to reveal even higher ambition levels in sustainable regional use of wood resources was a major task in the scenario analysis that was conducted as part of a SCBE research project on material flow analysis and logistics. The IS within the SCBE's case study system was assessed using the taxonomy of IS (Martin 2013, 2010) and the findings of the research team at the SCBE. Four different levels formed the basis of the scenario specification, as presented and proposed in Figure 1.

The first level focused on the overall holistic adoption of IS of wood-based stakeholders within the bioeconomy region. In other words, a focus on the geographic location where all IS activities (i.e., synergetic use of material streams) occur within two or more industrial parks to produce a regional wood-based product basket. The second level focused on the interaction and exchange among the different industrial parks, where wood-based material flows (i.e., raw materials and solid fuel by-products) are shared among wood panel producers, biorefinery plants, and polymer manufacturers. These are relevant for promoting integration among the various wood-based stakeholders, and hence supporting IS.¹ The third level focused on the activities within the individual industrial parks in terms of the intrasite exchange of heat and wood-based by-products. The fourth level then focused on the potential for actual synergetic integration options for by-product and waste revalorization among the individual production chains. At this level, material and energy flows are shared in addition to infrastructure (e.g., combined heating and power [CHP] plants).

Indeed, the future integration of bio-based production systems within existing industrial parks to promote the intrasite exchange of heat and by-products and to foster their transformation into eco-industrial parks is an important part of the system analysis for assessing IS within bioeconomy regions (Bezama 2018).

The case study system and the proposed methodology for extended life cycle assessment (LCA) scenarios differs from other studies, such as (Sokka et al. 2011; Eckelman and Chertow 2013), because the focus is much more on assessing the integrative function of lignocellulosic biorefineries for achieving regional environmental sustainability. It also assesses the exchanges of material flows between more central chemical industry sites and more remote sites of wood panel production.

During the planning stages of these bioeconomy regions, the options for sustainable synergies and collaboration between various industries producing bio-based polymers and engineered wood products (EWPs) need to be assessed, for example, to further prescreen whether an in-depth economic analysis of more integrated IS options should be conducted when the economic viability for integrated biorefinery concepts has already been evaluated (Nitzsche et al. 2016; Budzinski and Nitzsche 2016).

Therefore, the goal of the work presented in this paper was to identify the potential environmental advantages of the synergetic conversion and the coupled use of wood resources within these levels of IS in the SCBE.

Three scenarios were developed to investigate the options for IS within the SCBE (see the *System Boundaries and Scenarios for Increasing Levels of Industrial Symbiosis* section).

As outlined in Figure 2, the scenarios (Sc 1–Sc 3) were used for the following purposes:

1. To determine how the LCA results differ between conventionally organized value chains and a set of three future, increasingly synergetic, bio-based chains;
2. To identify environmental hotspots within the regional bioeconomy networks; and
3. To assess whether the environmental sustainability of the overall system increases in the integrated bioeconomy network.

The environmental performance of these symbiotic industrial relationships is therefore of particular strategic importance, as it determines the feasibility of such activities not only for the individual companies but also for the entire value-added networks in a wood-based bioeconomy region. This paper provides some answers to this important question.

Materials and Methods

Overview of the Life Cycle Assessment Methods

An LCA was used to evaluate the environmental impact of IS within SCBE's bio-based production systems. We first identified a future regional "bio-based product basket" and developed LCA models for all of SCBE's key individual products (Hildebrandt 2015; Hildebrandt et al. 2017b).

We then integrated these LCA models, including the IS options, to estimate the total environmental impact of the regional bioeconomy system. This was compared to the potential impact of a reference product basket composed of sample fossil-based counterparts without deploying IS scenarios. LCA modeling was performed using the software GaBi.

First, an LCA model was developed for each constituent product in the regional product basket. In successive steps, these individual LCA models were interlinked according to the demands and scope of the different scenarios, for example, with increasing degrees of supply chain integration and increasing degrees of substituting fossil-based auxiliaries, additives, and energy carriers. The implemented LCAs were cradle-to-gate.

The geographic scope of the LCA study was the region of Central Germany (CG) and Germany, with most of the foreground activities occurring in CG (see the *Data Collection, Data Aggregation, Data Quality, and Uncertainty Propagation* section).

A regional product basket approach was used in the LCA. This approach allowed suitable functional units to be defined for each of the individual products within the regional product basket (see the *Product Basket Approach, Functional Units, and Equality of Benefits* section).

The modeled LCAs were cradle-to-gate, covering the upstream chains of harvesting and feedstock transport (e.g.,

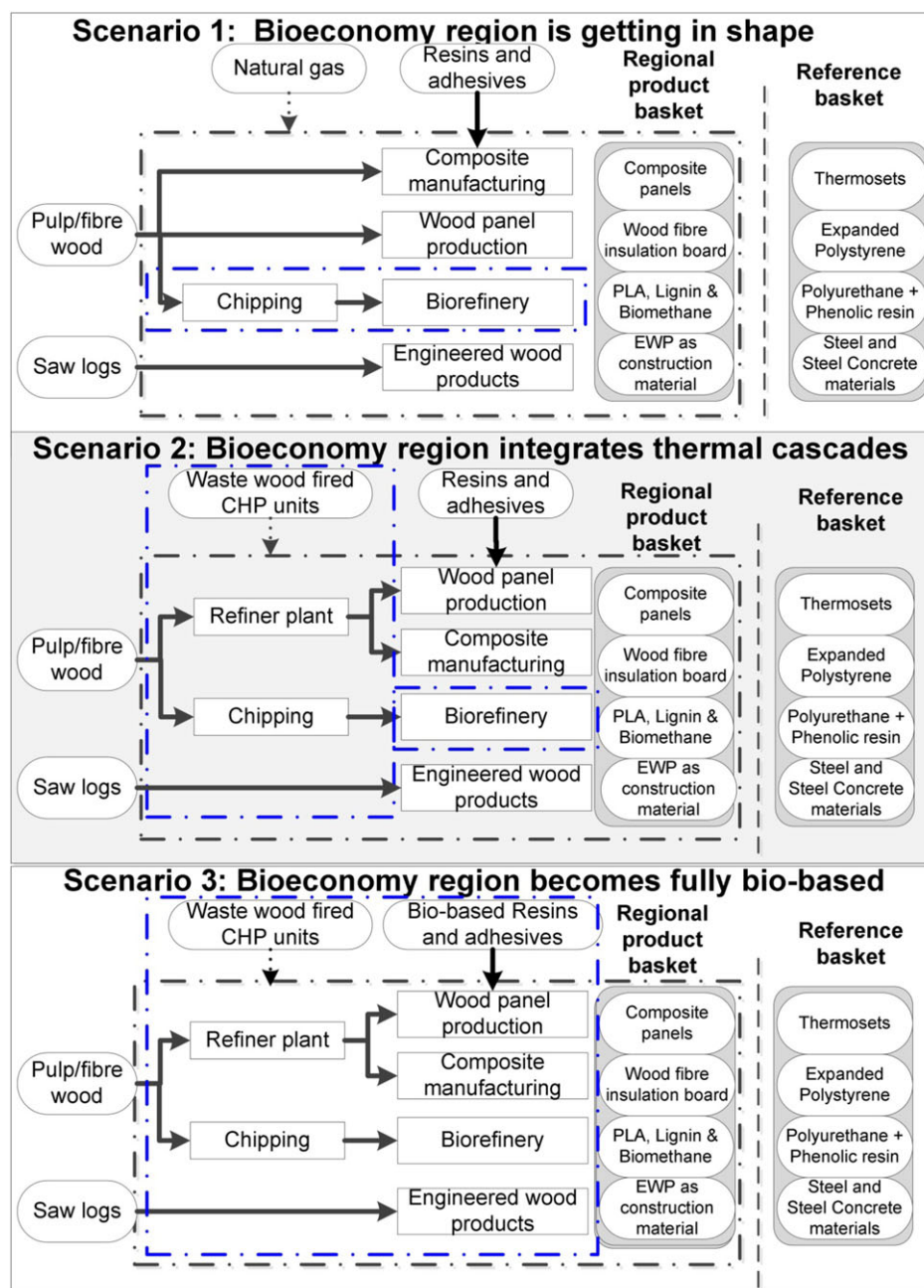


Figure 2 System boundaries and scenarios for the integration of value-added chains in the wood-based bioeconomy (Scenario Sc 1 was modified from Hildebrandt and colleagues [2017b]). (For each scenario the upstream and downstream processing paths where integration takes place are highlighted with blue and dashed lines.) See Table 2 for explanation of acronyms.

industrial pulp, wood fiber, and saw logs) and auxiliaries (e.g., industrial production of polylactic acid [PLA], fiber-reinforced composite materials, bio-based foams, and EWPs such as cross-laminated timber [CLT]).

The impact assessment of the LCA comparison between the bio-based product basket and its fossil-based counterpart used the CML 2013 method (impact assessment method developed by the Centrum voor Milieukunde [CML] Leiden, the Netherlands) (Institute of Environmental Sciences 2004; Guinée 2002).

The environmental sustainability of the different scenarios was compared across 11 of the CML impact categories. The CML method was chosen for the impact assessment because of its strength in comparing toxicity, which makes it exceptionally useful for comparing chemical products with each other. It was also chosen for its transparent and traceable midpoint indicators as impact categories. A feedstock demand of 630,000 tonnes absolute dry per annum (abs.dry/a) of wood resources was assumed to be the total annual production volume of the bioeconomy region.

System Boundaries and Scenarios for Increasing Levels of Industrial Symbiosis

The individual chains for producing wood-based products were based on the regional examples from the SCBE, demonstrating innovative wood processing technologies and efficient use of beech wood and hardwood products in the CG region (Hildebrandt 2015; Auer, et al. 2016; BMBF 2015; Hildebrandt et al. 2017b; Hildebrandt et al. 2017c).

From a product perspective, the focus of the SCBE was on the demonstration-scale production of bio-based products such as EWPs, composites, resins, and bio-based polymers made from beech wood.

The efficient and sustainable use of beech wood is strategically regarded as a cornerstone of the future wood-based bioeconomy in Germany. Beech wood is therefore seen as an increasingly relevant feedstock for the German wood industry and the integrated lignocellulosic biorefinery plants operated by members of the SCBE (Cluster Platform Germany Federal Ministry for Economic Affairs and Energy 2017).

To foster thermal integration across production plants within the regional industrial parks, options were assessed for the energetic cascading of waste wood, a by-product produced in the regional network. It can be used as a source of process energy, thus reducing dependency on fossil-sourced energy. The members of SCBE proposed options associated only within their individual context for IS (e.g., integration of biorefinery concepts or valorization of by-products), whereas the research team was tasked with structuring alternative options for IS from a systems perspective and according to their level of integration.

Three scenarios were established for regional product baskets with comparable equality of benefits in order to reflect and assess the possible options for the integration of value-added chains within the regional system boundaries of CG (Figure 2). These scenarios were based on the IS options proposed by members of the SCBE and structured by the systems analysis of the research team.

The three scenarios (Figure 2) were developed with the aim of determining the environmental benefits of gradually increasing the degrees of integration within the supply and conversion of the SCBE, a wood-based bioeconomy region. The levels of integration as well as the similarities and differences among the three scenarios are outlined in Table 1. It can be observed that while some core operations stay constant (e.g., wood pressing and drying, fiber drying, and laminate pressing processes), the individual scenarios differ in relation to the percentage of process energy derived from woody biomass, as well as the proportion of bio-based resins and adhesives in the individual scenarios (Table 1 and Figure 2).

The system boundary of the analysis is from cradle-to-gate. It includes the upstream chains of harvesting and transport of industrial round wood, the industrial production of bio-based platform chemicals, fiber-reinforced composite materials, bio-based foams, and EWPs. Furthermore, the cascading of waste

wood to supply process energy and the provision of regionally produced bio-based resins and adhesives are included in Sc 2 and Sc 3.

- Sc 1: “Getting in shape.” In this scenario, the bioeconomy region is getting in shape. A lignocellulosic biorefinery plant has been built in industrial park 2 and is producing bio-based polymers, lignin intermediates, and energy carriers. The introduction of this biorefinery means the sharing of wood-based feedstocks between the wood industry in industrial parks 1 and 3, and the chemical industry in industrial park 2. Thus, as part of the management and design of the value-added chains, we get the launch of vertical integration (e.g., novel sources for beech wood chips from industrial residues) and horizontal integration (e.g., novel options for intermediate business-to-business relations between biorefinery operators and composite manufacturers) by means of feedstock cascading. In this first scenario, infrastructures for supplying heat, steam, and electricity are not shared among production sites, and the production of heat and steam, for example, for fiber drying and thermal biorefinery processes, mainly relies on natural gas. Furthermore, the bio-based production of adhesives and resins is capable of replacing only 10% of the regional demand for resin binders and adhesives used in the production of composites, panel boards, and EWPs.
- Sc 2: “Integrating thermal cascades.” Here, the production pathways within the bioeconomy region are starting to integrate. The fuel demands required to operate the lignocellulosic biorefinery plant in industrial park 2, the fiber refining systems in industrial park 1, as well as fiber drying, wood drying, and panel pressing processes are fully covered by the incineration of waste wood, bark residues, and sawmill by-products as part of an intersite exchange (e.g., from industrial parks 1 and 3). Thus, natural gas is fully substituted thanks to the fulfillment of process energy demands through biomethane and thermal utilization of waste wood resources. Furthermore, a refinery plant to supply fiber is being jointly run by panel and composite manufacturers. This provides fiber for the production of composite materials and wood-based panels such as wood fiber insulation boards and medium-density fiberboards for both industrial parks 1 and 2. However, only 10% of the regional demand for resin binders and adhesives used in the production of composites, panel boards, and EWPs in all three industrial parks is being substituted by hardwood ethanol organosolv lignin (HEL).
- Sc 3: “Becoming fully bio-based.” In this scenario, there is increased interaction and a greater state of integration among the bioeconomy networks in the different industrial parks. Fossil-based resins and adhesives are fully replaced by bio-based adhesives and resins produced from lignin derivatives. Furthermore, all of the process energy required for production is bio-based, with all cooperation

Table 1 Three industrial symbiosis scenarios based on the level of value chain integration

Scenario name	Qualitative description	Level of integration	Resource base and substitution			
			Fuels		Materials	
			Fossil	Bio-based	Fossil	Bio-based
Getting in shape (Scenario 1—baseline scenario)	A biorefinery for producing bulk chemicals and commodities is commissioned and capacities for EWP are expanded.	Plants share wood-based feedstocks and bio-based chemicals.	80%	20%	5% (auxiliaries)	95%
Integrating thermal cascades (Scenario 2)	Thermal use of waste wood is integrated into the operation of the biorefinery and panel production. Start of joint operation of refinery plants for fiber production.	Plants share wood-based feedstocks, bio-based chemicals and bio-based process energy.	15% (upstream)	85%	5% (auxiliaries)	95%
Becoming fully bio-based (Scenario 3)	Biorefineries and the wood sector fully cooperate in producing bio-based materials.	System demands for fuel, auxiliaries, and materials are met by woody feedstock in the region.	10% (upstream)	90%	0%	100%

EWP = engineered wood product.

partners within the supply chains mutually sharing the benefits and all industrial parks covering a part of their fuel demand through an intersite exchange of residues from industrial wood. The thermal integration of waste wood-fired CHP units is fully developed within the individual industrial parks.

Product Basket Approach, Functional Units, and Equality of Benefits

The following production chains are the constituent production chains of the regional product basket and were thus integrated into the LCA framework according to the various scenarios modeled (Table 1 and Figure 2). When these production chains are organized into networks for sharing energy provision and intermediate products, these more integrated scenarios enable the assessment of the synergetic benefits of integrating the different regional production chains associated within the SCBE.

1. Production chains:

Production of resin, adhesives, and bio-based polymers such as PLA and HEL in a lignocellulosic biorefinery

- Production of EWPs such as laminated veneer lumber (LVL) beams and CLT walls
 - Production of composite materials reinforced with wood fibers
- Upstream chains: Harvesting and logistic chains for supplying industrial fiber, pulp wood, and saw logs from beech wood
 - Reference systems: LCA modeling of comparable fossil- and mineral-based reference production systems such as the production of expanded polystyrene (EPS), rigid polyurethane (PUR) boards, and concrete and steel products.

Data relating to the life cycle inventories and life cycle impact assessment of the three different production chains were based on studies by Auer and colleagues (2016), Hildebrandt (2015), Krombholz and colleagues (2015), and

Table 2 Definition of characteristic product and reference product portfolios for the wood-based bioeconomy region in the case study

Product category	Wood-based products	Reference products	Product applications	Functional units and equality of benefits	LCA models
Engineered wood product	CLT LVL GLB	RC SLB HE	Load-bearing walls, beams, stanchions	kg with same compressive strength, kg with same bending moment (MN*m)	Own models developed in cooperation with HIZ Südharz and Pollmeier (Pollmeier Massivholz GmbH & CoKG 2015)
Polymers and composites	E-PLA WFIB HEL resins NFRC	PF EPS PUR	Platform chemicals, insulation boards, construction materials, and interior construction	kg with same purity (polymer grade), m ² , m ³ with same overall heat transfer coefficient (W/m ² *K), m ³ with comparable flexural and compressive strength	Nitzsche and colleagues (2016), Kromholz and colleagues (2015), and own modeling of downstream processes
Energy carriers	BM HDL CIW SMBP	NG BC	Solid biofuels, heat and power	kg with same energy density (MJ or kWh), MJ or kWh feed into the high-voltage grid	Budzinski and colleagues (2016) and GaBi databases

BC = brown coal/lignite; BM = biomethane; CIW = chips from industrial wood; CLT = cross-laminated timber; E-PLA = expanded polylactic acid; EPS = expanded polystyrene; GLB = glued-laminated timber beam; HDL = hydrolysis lignin; HE = H-section beam, European; HEL = hardwood ethanol organosolv lignin; HIZ = Holzimpulszentrum (<http://www.holzimpulszentrum.de/>); kg = kilograms; kWh = kilowatt hour; LCA = life cycle assessment; LVL = laminated veneer lumber; m² = square meter; m³ = cubic meter; MJ = megajoule; NFRC = natural fiber reinforced composites; NG = natural gas (methane); PF = phenolic resin systems; PUR = rigid polyurethane; RC = reinforced concrete; SLB = sand-lime brick; SMBP = sawmill by-products/bark/sawdust; WFIB = wood fiber insulation boards.

Nitzsche and colleagues (2016) and elaborated for the case study region. The functional unit and equality of benefits for selected bio-based products and their references were as follows:

1. The functional unit of the overall system was set to an annual production volume of 430,000 tonnes per annum (t/a), which lies within the production capacity of the bioeconomy region.
The corresponding input capacities account for 630,000 tonnes per annum of wood resource.
2. Allocation by mass represents the volume share of the individual products within the regional product basket.
3. The wood-based products are compared to fossil-based reference products, offering equal benefits along the product use phase (see Table 2).

The primary mechanical and/or thermal properties of the building materials, polymer products, and biogenic fuels were quantified to determine the equality of benefits. An equivalent amount of fossil-based reference product with equal benefits was defined based on the regional production capacities for each individual wood-based product (Figure 3).

The most relevant properties for the selected bio-based products, which deliver the same or equal benefits per functional unit, are outlined in Table 2. Additional considerations to those outlined in Table 2 are the bio-product's estimated service lifetime, which must also be comparable. The functional units and equality of benefits work, for example, as follows: wood-fiber insulation boards (WFIBs) can have a thermal conductivity coefficient comparable to fossil-based EPS but require around three times the mass to achieve it (Figure 3). It can also be the other way around: CLT materials are more material efficient than the more conventional construction materials such as sand or sand-lime bricks (Figure 3). In terms of the energy carrier methane, biomethane is comparable to fossil-based methane and, as they have an equivalent energy density (i.e., Megajoule per tonne [MJ/t]), the annual production volume of 40,000 tonnes per annum of biomethane is comparable to the same amount of fossil-based methane (Figure 3).

Calculation of the Relative Advantage

The indicator results for the 11 CML impact categories (IR_c) were conventionally calculated by multiplying a specific characterization factor (CF) with the mass flow (m_s) of a particular

	Product basket - Bioeconomy Region	Reference Product Basket	
181,500	Hydrolysis lignin	Lignite	181,500
40,000	Biomethane	Methane	40,000
97,000	Expanded PLA	Expanded Polystyrene	97,000
4,500	Fibre reinforced composites and bio- foam	Polyurethane	5,400
80,000	Cross-laminated timber building systems	Reinforced concrete (load-bearing structure) & sand-lime brick building systems	520,000
22,500	Laminated veneer lumber (LVL) beams	Steel beams	31,500
4,000	Wood fibre insulation boards	Expanded polystyrene boards	1,400
Annual production volume [in t/a]		Reference production volume with equality of benefits [in t/a]	

Figure 3 Annual production volume of wood-based materials in the baseline case, Scenario 2 (modified from Hildebrandt and colleagues [2017b]).

substance (equation (1)). One example of a CF is the global warming potential (GWP) of a particular substance (Klöppfer and Grahl 2014; Guinée 2002).

$$IR_c = \sum_s CF \times m_s \quad (1)$$

The relative advantage or disadvantage of the environmental impact of the bio-based basket compared to the specific reference is then calculated by dividing the difference of the bio-based basket and the reference basket by the indicator results of the reference basket. This is expressed as a percentage (equation (2)). The following equation was used in the calculation:

$$\text{Rel. advantage} = \frac{IR_{c-\text{Bio-based Basket}} - IR_{c-\text{Reference}}}{IR_{c-\text{Reference}}} \text{ in [\%]} \quad (2)$$

Data Collection, Data Aggregation, Data Quality, and Uncertainty Propagation

The inventory analysis for the individual products and innovative processes was conducted in cooperation with material science research partners (e.g., wood technology and polymer sciences) and with industrial partners (Krombholz et al. 2015; Hildebrandt 2015; Nitzsche et al. 2016). It was done in collaboration with SCBE's various industrial partners (Krombholz

et al. 2015; Hildebrandt 2015; Budzinski and Nitzsche 2016; Nitzsche et al. 2016; Auer et al. 2016).

In addition to the specific data from regional research and industry partners, datasets from Ecoinvent and PE International/ThinkStep were included in the assessment (Table 3). The final LCA model aggregates the LCA models for the individual process chains and for the reference products.

When tracing back the contribution analysis, the allocation of impacts in the assessed multi-output product systems such as biorefinery production was done by mass. Allocating by price does not seem reasonable because high-priced polymers such as PLA would then outweigh low-priced commodities such as the hydrolysis lignin.

The life cycle inventory data cannot be presented in this paper because of the sensitivity of some of the inventory datasets for the biorefinery process chains and the wood panel and composites products, which make up a large fraction of output products, and due to the confidentiality of these datasets.

The data quality and the associated variances and uncertainties are another important aspect to consider when assessing future product systems and when evidence-based findings are consulted for more environmentally sound decision making. Variances and uncertainties can occur in different process

Table 3 Inventory data, unit processes, and associated datasets used in the life cycle assessment modeling

<i>Reference processes</i>	<i>Database or source</i>	<i>Bio-based processes</i>	<i>Database or source</i>
Production of polystyrene boards, expanded, at factory	Kellenberger (2007)	- Wood-fiber insulation boards - EU-27: Methylene diphenyl diisocyanate ([p]MDI) ISOPA - Transport of wood chips - Transport of paraffin - Refinery process - Fiber drying - Curing of adhesive - Cutting and packaging <e-ep> - Diammonium phosphate, as P2O5 - RER: Wood chips, hard wood - RER: Wood chips, soft wood - RER: Paraffin, at factory gate	- GaBi database, PE International (2013) - Communications and site visits with HIZ and FH Rosenheim - Own modeling - Werner (2007) - Zah (2007)
Production of steel profiles GLO: steel profiles, CN: Sand-lime bricks CN: Pumping of concrete 1m ³ CN: Transport concrete C20/25 GLO: Reinforced concrete	(Worldsteel 2011) PE GaBi database (2013) PE GaBi database (2013) PE GaBi database (2013) PE GaBi database (2013), Worldsteel (2011)	- Cross-laminated wall elements - CN: Transport, truck - Natio: Gluing <e-ep> - Natio: CLT press <e-ep> - RER: Diesel, at regional storage - RER: Melamine-urea-formaldehyde resin, at factory gate - RER: Sawn timber, hardwood, planed, chamber dried, - RER: Sawn timber, softwood, planed, chamber dried,	- Own modeling - Communications and site visits with HIZ and FH Rosenheim - Althaus (2007) - Werner (2007)
RER: Polystyrene boards, expanded, at factory gate	Kellenberger (2007)	- Biorefinery concept 3.2 (cradle- to-gate, aggregated), lactic acid <e-ep> - Catalytic conversion of lactic acid - Polymerization of PLA <e-ep> - Production of PLA, granulates - Foam expansion and molding	- Budzinski and Nitzsche (2016) - Nordegraaf (2013); Synbra (2015); Synbra Technology (2015) - Groot and Borén (2010)

(Continued)

Table 3 Continued

<i>Reference processes</i>	<i>Database or source</i>	<i>Bio-based processes</i>	<i>Database or source</i>
DE: Lignite briquettes	Ecoinvent (1988), PE GaBi database (2011)	Biorefinery concept 3.2 (cradle-to-gate, aggregated), hydrolysis lignin	- Budzinski and Nitzsche (2016)
DE: Methane PE	PE GaBi database (2011)	Biorefinery concept 3.2 (cradle-to-gate, aggregated),	- Budzinski and Nitzsche (2016)
RER: Polyurethane—hard foam (PUR)	PlasticsEurope, GaBi database (2006)	- Biorefinery concept 3.2 (cradle-to-gate, aggregated), lignin - FR: Flax long fibers PE - IN: Kenaf fibers, at farm - Natio: Thermal press <e-ep> - Natio: Prepreg production - Foaming of NFK-PU elements - OCE: Transport, freight overseas	- Budzinski and colleagues (2016) - Communications with Fraunhofer IWM (2014) - GaBi database (2011) - Althaus (2007) - Unkelbach and colleagues (2009), Pufky-Heinrich and colleagues (2016)

CLT = cross-laminated timber; CN = China; DE = Germany; FR = France; GLO = Global; IN = India; NFK-PU = Natural fibre reinforced composite with polyurethane core layer; OCE = Oceanic; PE = Company PE International/ThinkStep; PLA = polylactic acid; RER = Europe.

stages and scales during the inventory phase of LCA, as well as during impact estimation.

The calculation of variances and uncertainties that must be considered include:

1. gaps in data quality between lab, demo-scale, and industrial scale processes;
2. possible future options for impact mitigation;
3. the impact mitigation already achieved and not accounted for in older datasets;
4. options for impact mitigation in the upstream and downstream chains; and
5. the general range of variation of process coefficients in generic process models.

The uncertainties relating to the key modeling carried out in this paper are outlined in Table 4.

The propagation of uncertainty is illustrated by the error bars presented in Figure 4, which also reflects variances between different sources of LCA databases when they exhibit significant differences and when different sources were available for comparison (e.g., for steel and lignite briquettes).

Results

The 11 impact categories of CML 2013 were evaluated with regard to the individual production chains and the regional product basket of each scenario. The inventory analysis includes highly detailed conversion coefficients, consumption factors,

and emission factors but is presented in a more aggregated manner because some of the process parameters are confidential. LCA indicators related to land use were not considered in the assessment because the approach implied a cap-assessed production system. A regional feedstock supply of 630,000 tonnes per annum, covered by the regional catchment area, was used, as it is the most realistic assumption for the initial transformation phase toward becoming a bioeconomy region. On a midterm perspective, these feedstock demands are expected to stay within the carrying capacity limits of the regional net biome production (Husmann et al. 2016).

Analysis of Environmental Impact

The results of the life cycle impact assessment (LCIA) for the regional bio-based product basket of each scenario are compared with the fossil-based reference basket (Sc 1) to assess their relative advantage.

The results for Sc 1 indicate that the introduction of bio-based feedstocks into the system reduces the environmental impacts for seven out of eleven impact categories (Figure 4). Of the four categories that show less favorable outcomes for bio-based production, two categories—namely, acidification potential (AP) and ozone depletion potential (ODP)—have a 200% higher environmental burden than the conventional fossil-based reference. Photochemical ozone creation is still relatively higher, while the human toxicity potential (HTP) has decreased. It is still in the range of basic uncertainties with a difference of less than 5%. Improvement in the

Table 4 Characterization of variances and uncertainties¹

Inventory data for all processes representing the hot spots of energy and resource intensity			Impact categories and their average uncertainty levels calculated for the example of Scenario 2		
Unit processes and auxiliary material demands	Process coefficients specified in inventory analysis	Variances/degrees of their uncertainty			
Resin production and associated upstream processes	- Resin demand - Eco-efficiency of urea formaldehyde production	±5%	ADP	−12%	14%
			ODP,	−4%	5%
			EP	−8%	10%
			TETP	−8%	10%
			POCP	−5%	6%
			AP	−7%	9%
			FAETP	−12%	15%
			HTTP	−10%	12%
			GWP 100 a	4%	−5%
			ADP fossil	−7%	15%
Process heat for drying, pressing and fractionation	- Cumulated heat demand [kWh/tonne]	± 75 kWh (refiner) ± 15 kWh (presses) + 120 kWh (fraction.) − 65 kWh (fraction.) per tonne of product	MAETP	−11%	14%
Electricity demand	- Cumulated electricity demand	±7.5 %			

Composition of the product mix

Uncertainties were calculated using scenario analysis and the parameter variation function in the GaBi Software

ADP elements = abiotic depletion of elements (kg Sb-equiv.); ADP fossil = abiotic depletion of fossils (MJ); AP = acidification potential (kg SO₂ equiv.); EP = eutrophication potential (kg phosphate equiv.); FAETP = freshwater aquatic ecotoxicity (kg DCB equiv.); GWP 100 a = global warming potential (kg CO₂ equiv.); HTTP = human toxicity potential (kg DCB equiv.); kWh = kilowatt hours; MAETP = marine aquatic ecotoxicity (kg DCB equiv.); POCP = photochemical ozone creation (kg ethene equiv.); TETP = terrestrial ecotoxicity potential 9 kg DCB equiv.).

environmental impacts demonstrates the benefits of Sc 1: “Getting in Shape.”

The results for Sc 2 indicate that the use of waste wood to supply process energy and steam contributes to lowering even further the eutrophication potential and the marine aquatic ecotoxicity. The highest positive impact can be observed for the human toxicity potential, which has a relative advantage in Sc 2. However, the AP and the ODP increase further. This is related to the integration of a waste wood CHP into the supply chains for heat and power. The global warming potential (GWP) of the multiproduct system remains almost constant. This is related to the fact that the greenhouse gas mitigation potentials are low when natural gas is replaced by waste wood-based fuels in Sc 2. In this case, the biogenic carbon stored in the wood products is released back into the atmosphere. Furthermore, the natural gas used in Sc 1 is considered to be the most climate-efficient fossil fuel, with low carbon dioxide emissions.

The downstream processing of bio-based resins and the full substitution of energy was modeled in Sc 3. The results of the environmental assessment reveal that there are no further benefits in relation to the impact categories FAETP (freshwater aquatic ecotoxicity), HTTP (human toxicity potential), GWP

100 a (global warming potential), ADP fossil (abiotic depletion of fossils), and MAETP (marine aquatic ecotoxicity). It also shows that the POCP (photochemical ozone creation potential) significantly increases for the bio-based production systems due to additional demands for process energy in the downstream processing of 60,000 tonnes per annum of lignin per annum.

Discussion

The environmental performance of IS relationships within a bioeconomy region is a question of particular strategic importance as it determines the feasibility of such activities for the individual companies as well as for the entire value-added networks within a wood-based bioeconomy region. The modeling results presented in this paper, which are based on a comparison of scenarios, help to provide an answer to this question. The three scenarios helped to identify the environmental benefits of (1) integrating biorefineries, (2) energetic cascading of waste wood, and (3) downstream processing of lignin compounds and regional consumption of high-value-added bio-based products in the business-to-business relationships of regional wood-based industry networks.

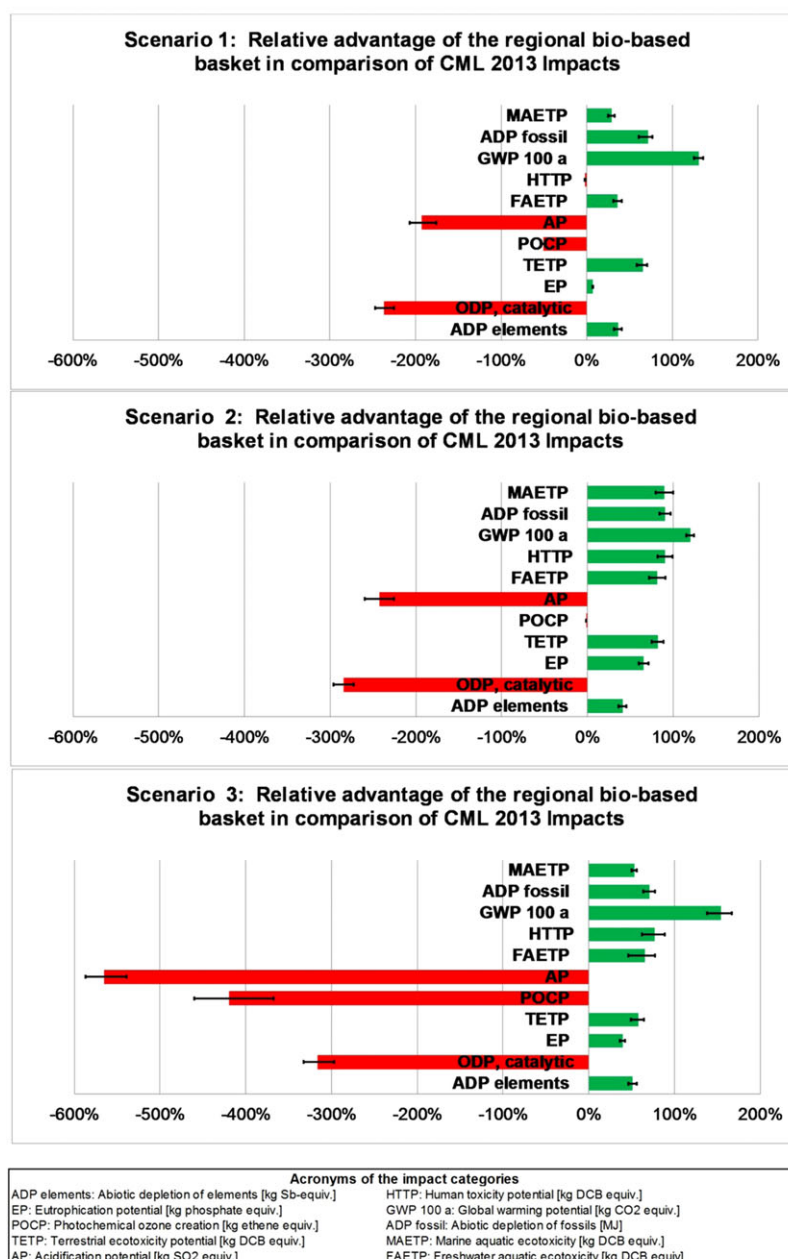


Figure 4 Scenarios 1–3 life cycle impact assessment comparison of the relative advantage of bio-based products in minimizing the environmental impacts over fossil-based references (green highlights benefits for the bio-based products, and red highlights benefits for the fossil-based references) (Scenario 2 was modified from Hildebrandt and colleagues [2017b]). (See Table 4 for explanation of acronyms.)

A comparison of the environmental impacts related to the production of bio-based products and the production of corresponding fossil reference products with equal benefits revealed a significant divergence in the results of the impact assessment.

The production of a mixed product portfolio consisting of wood-based and a high proportion of fossil-based products, as represented by the reference scenario (see the *Traditional Production Base* section within the supporting information available on the Journal's website), had a poorer environmental performance compared to bio-based references in 9 of 11 impact categories.

In contrast, integrating biorefinery plants into the wood industry system to develop future bioeconomy networks, as represented by Sc 1, contributes to reducing environmental impacts in 7 of 11 impact categories. Only for AP and ODP did the fossil-based counterparts exhibit significantly less impact along their product life cycle. This was mainly related to the integration of a waste wood-fired CHP using treated waste wood fractions into the supply chains for heat and power. Furthermore, in Sc 3, the additional demand for process energy in the downstream processing of 60,000 tonnes per annum of lignin per annum outweighs some of the benefits that were obtained

by substituting fossil-based phenolic resins in the first place. The results show that there is a 75% to 80% reduction in the impact on global warming and the eco-toxicity potentials with the implementation of a future integrated bioeconomy network (Sc 3 and Sc 4).

One key hot spot is the AP, which is a common environmental issue associated with biomass production and conversion and is mainly attributed to the gaseous emissions of wood drying and biomass-fired combustion facilities. On the other hand, major improvements can be made by using waste wood to supply heat and power, which contributes significantly to lowering the environmental impact of six impact categories, namely eutrophication potential, marine aquatic eco-toxicity, photochemical ozone creation, freshwater aquatic eco-toxicity, human toxicity potential, and abiotic depletion of fossils. Therefore, Sc 2 can be referred to as a blueprint on how to integrate the cascading use of waste wood, coupled with the use of biorefinery products, to contribute to a substantial reduction in negative environmental impacts by replacing fossil resources. Substituting process energy reduces overall environmental impacts by 10%. In this regard, to incentivize companies to invest in efficiently shared infrastructures for heat supply from waste wood CHPs, policy makers should take into greater consideration the integration of industrial heat from woody residues over installations primarily driven by electricity when striving for coherent renewable energy policies, waste laws, and the bioeconomy domains.

When assessing absolute impacts of the production networks in the bioeconomy region (see the *Exemplary Contribution Analysis* section within the supporting information on the Web), we observe that the production of bio-based polymers such as PLA and the upstream chains of adhesives used for EWP production show significantly higher impacts for all CML 2013 categories than the production of lignin compounds and the production of EWPs excluding the impact of the utilized adhesives.

Therefore, the substitution of fossil-based adhesives and resins with lignin compounds is a potential corridor of action that regional experts and practitioners working in cluster initiatives should integrate into their future strategies. In particular, they should consider further R&D efforts in bringing to technological maturity low-energy enzymatic processes for lignin depolymerization and functionalization.

Furthermore, more competitive and profiled composite product groups should be developed in the first phases of market introduction and capacity upscaling so that large volumes of phenolic resins can be replaced by polymeric lignin substitutes.

Enhanced emission control and mitigation of environmental impacts in waste wood-fired CHP units will increase in importance within the industrial parks of the wood panel industry and the chemical industry, as utilization and modification of beech wood resources for material use applications increases.

For LCA practitioners, this also requires keeping inventories and impact assessments up to date with regard to these types of heat and power technologies to enhance the environmental benefits of close-to-industry waste wood use in the coming decades for more sustainable thermal integration within industrial parks.

In this regard, this study can serve as an orienting scenario framework in the future when industrial ecology management in bioeconomy clusters becomes more widely adopted and can provide valuable benchmarking figures when further accompanying LCA studies on wood-based bioeconomy systems are conducted.

In terms of the major limitations of this study data availability, data quality and technological maturity need to be mentioned. In general, this study had a more robust database on upscaling the production of EWPs due to better data availability, data quality, and technological maturity. The database for the fermentation processes of wood-based sugars within biorefineries had a higher degree of uncertainties with regard to their final conversion efficiencies in industrial scale applications. Even though more biotechnological conversion processes were studied in the SCBE, such as acetone-butanol-ethanol fermentation, they were not addressed in the LCA modeling due to their lower technological maturity or because of restrictions in data availability.

Conclusions

The applied LCA methodology, which aggregated and integrated LCA models for individual bio-based conversion and production chains, has proven to be a powerful tool in assessing IS within regional bio-based production networks from a higher level of process integration and data aggregation.

The scenario results show that, in most cases, the bioeconomy network outperformed the fossil-based production system, mitigating environmental impacts by 25% to 130%. Advancements are made between the reference scenario and Sc 1, with the integration of a biorefinery into existing and innovative EWP and composite production systems, which already produces significant environmental advantages. Substituting fossil fuels, such as lignite and methane, with hydrolysis lignin, bark residues, and waste wood is also highly relevant for realizing further environmental benefits.

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Note

1. In terms of the status quo, this study excludes conclusive interpretations of whether or not existing industrial parks in the case study

region, such as Leuna and the wood industry zones around Rottleberode, already meet the requirements of being eco-industrial parks. However, the future scenarios on thermal integration and integration of biorefinery chains can be regarded as effective activities for promoting the transformation of existing industrial parks towards future eco-industrial parks.

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Supporting Information

Supporting information is linked to this article on the *JIE* website:

Supporting Information S1: This supporting information provides insights into the inventory analysis for specifying material flows and unit processes of the bioeconomy scenarios and shows resource demands and availabilities for modeling the options for thermal integration and by-product valorization. Furthermore, an exemplary contribution analysis for Scenario 2 from Thinkstep GaBi Analysis is included, and a description and assessment of the reference scenario “Traditional production base” supports the understanding of the conventional production systems.